

fong-spivak-seven-sketches: image evidence

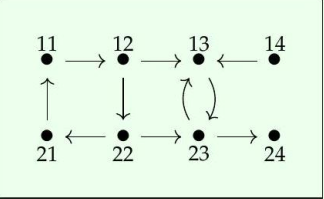
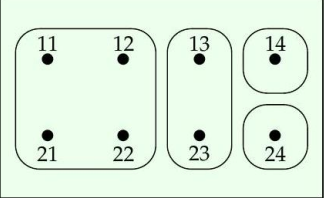
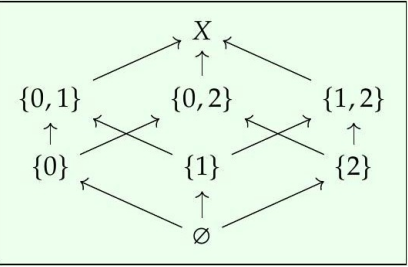
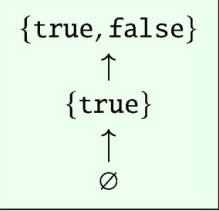
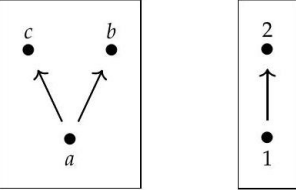
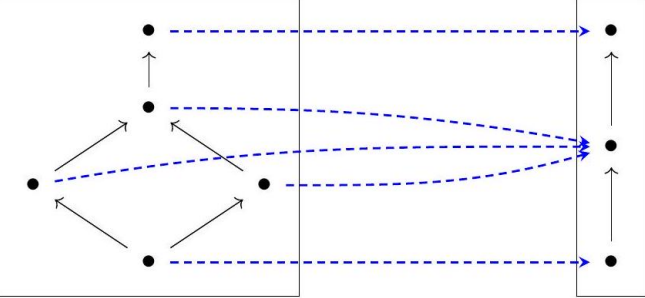
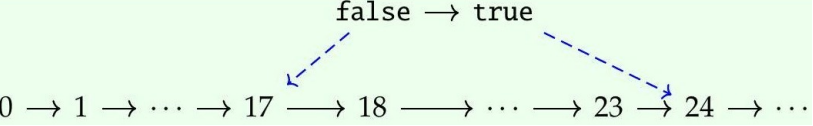
Image regions

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_4 DIA_0002	4	—	—	—	
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fong-spivak-seven-sketches_15 DIA_0004	15	—	—	—	
fong-spivak-seven-sketches_15 DIA_0005	15	—	—	—	
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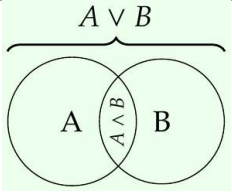
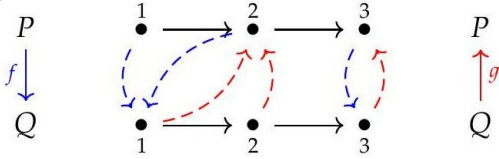
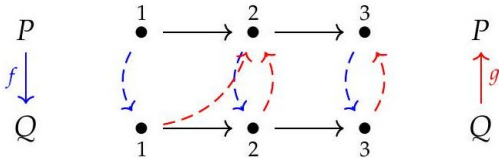
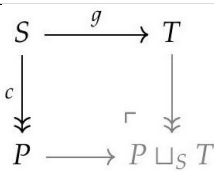
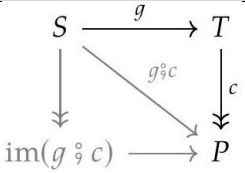
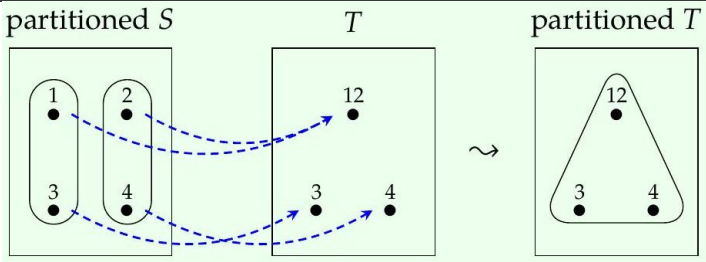
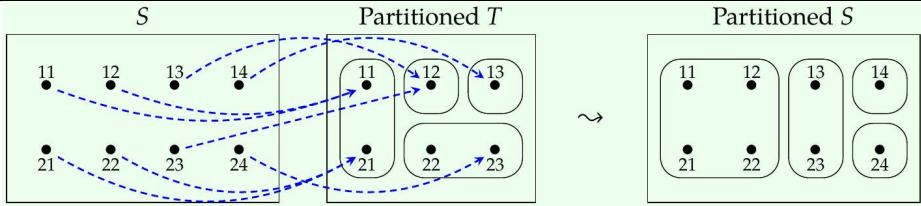
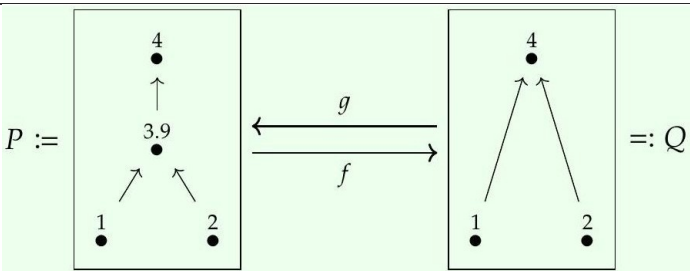
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fong-spivak-seven-sketches_16 DIA_0011	16	—	—	—	
fong-spivak-seven-sketches_16 DIA_0012	16	—	—	—	
fong-spivak-seven-sketches_16 DIA_0013	16	—	—	—	
fong-spivak-seven-sketches_17 DIA_0014	17	—	—	—	
fong-spivak-seven-sketches_17 DIA_0015	17	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_0016 DIA_0016	18	—	—	—	true ↑ false
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fong-spivak-seven-sketches_0018 DIA_0018	23	—	—	—	$\underline{3} \rightarrow \underline{3}$
fong-spivak-seven-sketches_0019 DIA_0019	23	—	—	—	surjective func $\underline{3} \twoheadrightarrow \underline{2}$
fong-spivak-seven-sketches_0020 DIA_0020	23	—	—	—	injective function $\underline{2} \hookrightarrow \underline{3}$
fong-spivak-seven-sketches_0021 DIA_0021	23	—	—	—	$\underline{2} \hookrightarrow \underline{3}$
fong-spivak-seven-sketches_0022 DIA_0022	23	—	—	—	bijjective funct $\underline{3} \xrightarrow{\cong} \underline{3}$
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_23 DIA_0027	23	—	—	—	
fong-spivak-seven-sketches_24 DIA_0028	24	—	—	—	
fong-spivak-seven-sketches_24 DIA_0029	24	—	—	—	
fong-spivak-seven-sketches_24 DIA_0030	24	—	—	—	
fong-spivak-seven-sketches_25 DIA_0031	25	—	—	—	
fong-spivak-seven-sketches_25 DIA_0032	25	—	—	—	
fong-spivak-seven-sketches_26 DIA_0033	26	—	—	—	
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fong-spivak-seven-sketches_0038	29	—	—	—	
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_0055	47	—	—	—	$\emptyset \longrightarrow \{(1,1)\}$
fong-spivak-seven-sketches_0056	52	—	—	—	
fong-spivak-seven-sketches_0057	52	—	—	—	
fong-spivak-seven-sketches_0058	55	—	—	—	$x \leq y$
fong-spivak-seven-sketches_0059	55	—	—	—	x_0
fong-spivak-seven-sketches_0060	55	—	—	—	$x_0 \leq x_1$
fong-spivak-seven-sketches_0061	55	—	—	—	$x_0 \leq x_1 \leq x_2$
fong-spivak-seven-sketches_0062	55	—	—	—	$x_0 \leq x_1 \leq x_2 \leq x_3$
fong-spivak-seven-sketches_0063	56	—	—	—	
fong-spivak-seven-sketches_0064	56	—	—	—	
fong-spivak-seven-sketches_0065	56	—	—	—	

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fong-spivak-seven-sketches_0067 DIA_0067	56	—	—	—	
fong-spivak-seven-sketches_0068 DIA_0068	56	—	—	—	$\frac{x}{y}$
fong-spivak-seven-sketches_0069 DIA_0069	56	—	—	—	$x \otimes y$
fong-spivak-seven-sketches_0070 DIA_0070	57	—	—	—	$\frac{\quad}{I} \quad \text{nothing}$
fong-spivak-seven-sketches_0071 DIA_0071	57	—	—	—	
fong-spivak-seven-sketches_0072 DIA_0072	57	—	—	—	$\frac{3.14}{3.14} \quad \frac{\quad}{-1} \quad \left. \frac{3.14}{-1} \right\} = \frac{\quad}{2.14} \quad \frac{\quad}{0} = \text{nothing}$
fong-spivak-seven-sketches_0073 DIA_0073	57	—	—	—	
fong-spivak-seven-sketches_0074 DIA_0074	57	—	—	—	$\frac{\quad}{x}$
fong-spivak-seven-sketches_0075 DIA_0075	57	—	—	—	
fong-spivak-seven-sketches_0076 DIA_0076	58	—	—	—	
fong-spivak-seven-sketches_0077 DIA_0077	58	—	—	—	
fong-spivak-seven-sketches_0078 DIA_0078	58	—	—	—	$\frac{\text{nothing}}{x} \quad \frac{\quad}{x}$
fong-spivak-seven-sketches_0079 DIA_0079	58	—	—	—	
fong-spivak-seven-sketches_0080 DIA_0080	58	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_59 DIA_0082	59	—	—	—	
fong-spivak-seven-sketches_61 DIA_0083	61	—	—	—	
fong-spivak-seven-sketches_62 DIA_0084	62	—	—	—	
fong-spivak-seven-sketches_62 DIA_0085	62	—	—	—	
fong-spivak-seven-sketches_63 DIA_0086	63	—	—	—	
fong-spivak-seven-sketches_63 DIA_0087	63	—	—	—	
fong-spivak-seven-sketches_63 DIA_0088	63	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_2 DIA_0090	70	—	—	—	
fong-spivak-seven-sketches_3 DIA_0091	72	—	—	—	
fong-spivak-seven-sketches_4 DIA_0092	74	—	—	—	
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fong-spivak-seven-sketches_0097	90	—	—	—	
fong-spivak-seven-sketches_0098	91	—	—	—	
fong-spivak-seven-sketches_0099	91	—	—	—	<pre> schema mySchema = { nodes Employee, Department; attributes DName : Department -> string, FName : Employee -> string; arrows Mngr : Employee -> Employee, WorksIn : Employee -> Department, Secr : Department -> Employee; equations Department.Secr.WorksIn = Department, Employee.Mngr.WorksIn = Employee.WorksIn; } </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_0100	92	—	—	—	$A :=$ $=: B$
fong-spivak-seven-sketches_0101	95	—	—	—	
fong-spivak-seven-sketches_0102	96	—	—	—	$\text{Free_square} :=$
fong-spivak-seven-sketches_0103	96	—	—	—	$\text{Comm_square} :=$
fong-spivak-seven-sketches_0104	96	—	—	—	
fong-spivak-seven-sketches_0105	97	—	—	—	
fong-spivak-seven-sketches_0106	97	—	—	—	

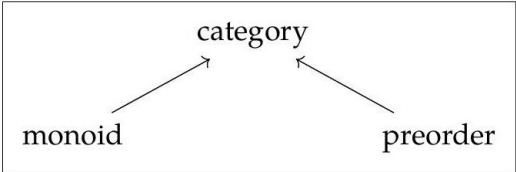
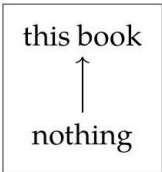
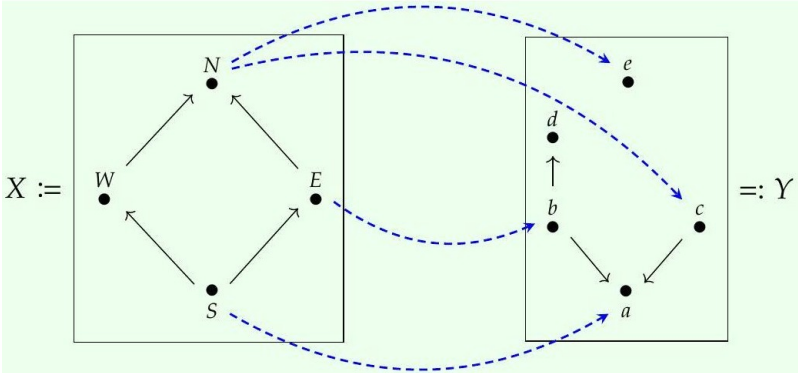
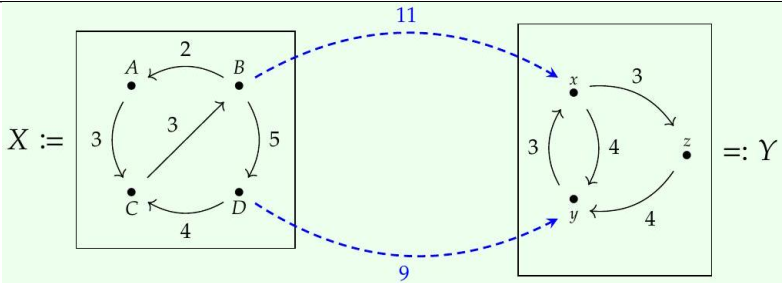
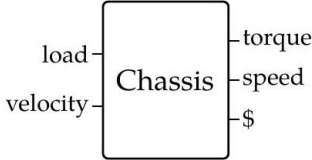
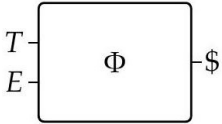
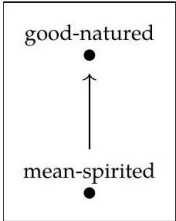
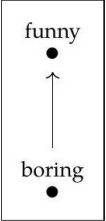
Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_0107	97	—	—	—	
fong-spivak-seven-sketches_0108	98	—	—	—	$G_1 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \bullet \xrightarrow{g} \bullet \\ \hline \end{array} \quad G_2 = \begin{array}{ c } \hline \bullet \\ \hline \bullet \\ \hline \end{array} \quad G_3 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \bullet \xrightarrow{i} \bullet \\ \hline \bullet \xrightarrow{g} \bullet \\ \hline \bullet \xrightarrow{h} \bullet \\ \hline \end{array} \quad G_4 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \bullet \xrightarrow{g} \bullet \\ \hline \bullet \xrightarrow{h} \bullet \\ \hline \end{array} \diamond$
fong-spivak-seven-sketches_0109	101	—	—	—	
fong-spivak-seven-sketches_0110	101	—	—	—	
fong-spivak-seven-sketches_0111	102	—	—	—	Planet of Sol
fong-spivak-seven-sketches_0112	102	—	—	—	Prime number
fong-spivak-seven-sketches_0113	102	—	—	—	Flying pig
fong-spivak-seven-sketches_0114	102	—	—	—	Beatle $\xrightarrow{\text{Played}}$ Rock-and-roll instrument
fong-spivak-seven-sketches_0115	103	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_103 DIA_0116	103	—	—	—	
fong-spivak-seven-sketches_103 DIA_0117	103	—	—	—	
fong-spivak-seven-sketches_104 DIA_0118	104	—	—	—	Free_square :=
fong-spivak-seven-sketches_104 DIA_0119	104	—	—	—	Comm_square :=
fong-spivak-seven-sketches_106 DIA_0120	106	—	—	—	1.
fong-spivak-seven-sketches_106 DIA_0121	106	—	—	—	
fong-spivak-seven-sketches_106 DIA_0122	106	—	—	—	$f \circ g = f \circ h$
fong-spivak-seven-sketches_107 DIA_0123	107	—	—	—	
fong-spivak-seven-sketches_107 DIA_0124	107	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_108 DIA_0125	108	—	—	—	
fong-spivak-seven-sketches_109 DIA_0126	109	—	—	—	$\mathbf{Gr} := \begin{array}{c} \text{Arrow} \begin{array}{c} \xrightarrow{\text{source}} \\ \xleftarrow{\text{target}} \end{array} \text{Vertex} \\ \text{no equations} \end{array}$
fong-spivak-seven-sketches_110 DIA_0127	110	—	—	—	$\begin{array}{ccc} G(\text{Arrow}) & \xrightarrow{\alpha_{\text{Arrow}}} & H(\text{Arrow}) \\ G(\text{source}) \downarrow & & \downarrow H(\text{source}) \\ G(\text{Vertex}) & \xrightarrow{\alpha_{\text{Vertex}}} & H(\text{Vertex}) \end{array}$
fong-spivak-seven-sketches_110 DIA_0128	110	—	—	—	$\begin{array}{ccc} G(\text{Arrow}) & \xrightarrow{\alpha_{\text{Arrow}}} & H(\text{Arrow}) \\ G(\text{target}) \downarrow & & \downarrow H(\text{target}) \\ G(\text{Vertex}) & \xrightarrow{\alpha_{\text{Vertex}}} & H(\text{Vertex}) \end{array}$
fong-spivak-seven-sketches_112 DIA_0129	112	—	—	—	
fong-spivak-seven-sketches_113 DIA_0130	113	—	—	—	
fong-spivak-seven-sketches_114 DIA_0131	114	—	—	—	$\mathcal{C} \xrightarrow{F} \mathcal{D} \begin{array}{c} \xrightarrow{I} \\ \Downarrow \alpha \\ \xrightarrow{J} \end{array} \mathbf{Set} \quad \leadsto \quad \mathcal{C} \begin{array}{c} \xrightarrow{F \circ I} \\ \Downarrow \alpha_F \\ \xrightarrow{F \circ J} \end{array} \mathbf{Set}$
fong-spivak-seven-sketches_115 DIA_0132	115	—	—	—	$\begin{array}{ccc} \mathcal{C}(c, Rd) & \xrightarrow{\alpha_{c,d}} & \mathcal{D}(Lc, d) \\ \mathcal{C}(f, Rg) \downarrow & & \downarrow \mathcal{D}(Lf, g) \\ \mathcal{C}(c', Rd') & \xrightarrow{\alpha_{c',d'}} & \mathcal{D}(Lc', d') \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_116 DIA_0133	116	—	—	—	
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fong-spivak-seven-sketches_121 DIA_0135	121	—	—	—	
fong-spivak-seven-sketches_121 DIA_0136	121	—	—	—	
fong-spivak-seven-sketches_122 DIA_0137	122	—	—	—	
fong-spivak-seven-sketches_123 DIA_0138	123	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_0140	123	—	—	—	
fong-spivak-seven-sketches_0141	125	—	—	—	
fong-spivak-seven-sketches_0142	125	—	—	—	
fong-spivak-seven-sketches_0143	125	—	—	—	
fong-spivak-seven-sketches_0144	130	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches DIA_0145	132	—	—	—	$x :=$  $y :=$ 
fong-spivak-seven-sketches DIA_0146	134	—	—	—	
fong-spivak-seven-sketches DIA_0147	135	—	—	—	
fong-spivak-seven-sketches DIA_0148	136	—	—	—	
fong-spivak-seven-sketches DIA_0149	136	—	—	—	
fong-spivak-seven-sketches DIA_0150	136	—	—	—	$T :=$ 
fong-spivak-seven-sketches DIA_0151	136	—	—	—	$E :=$ 

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches DIA_0152	136	—	—	—	
fong-spivak-seven-sketches DIA_0153	137	—	—	—	
fong-spivak-seven-sketches DIA_0154	138	—	—	—	
fong-spivak-seven-sketches DIA_0155	139	—	—	—	
fong-spivak-seven-sketches DIA_0156	140	—	—	—	
fong-spivak-seven-sketches DIA_0157	141	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image																									
fong-spivak-seven-sketches_141 DIA_0158	141	—	—	—																										
fong-spivak-seven-sketches_142 DIA_0159	142	—	—	—																										
fong-spivak-seven-sketches_144 DIA_0160	144	—	—	—																										
fong-spivak-seven-sketches_144 DIA_0161	144	—	—	—	<table><tr><th>Col(Φ)</th><th>A</th><th>B</th><th>x</th><th>y</th></tr><tr><td>A</td><td>0</td><td>2</td><td>5</td><td>8</td></tr><tr><td>B</td><td>∞</td><td>0</td><td>∞</td><td>∞</td></tr><tr><td>x</td><td>0</td><td>0</td><td>0</td><td>3</td></tr><tr><td>y</td><td>0</td><td>0</td><td>4</td><td>0</td></tr></table>	Col(Φ)	A	B	x	y	A	0	2	5	8	B	∞	0	∞	∞	x	0	0	0	3	y	0	0	4	0
Col(Φ)	A	B	x	y																										
A	0	2	5	8																										
B	∞	0	∞	∞																										
x	0	0	0	3																										
y	0	0	4	0																										
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches DIA_0178	152	—	—	—	
fong-spivak-seven-sketches DIA_0179	152	—	—	—	
fong-spivak-seven-sketches DIA_0180	153	—	—	—	$ \begin{array}{ccc} c & \xlongequal{\quad} & c \\ \cong \downarrow & & \uparrow \cong \\ c \otimes I & & I \otimes c \\ c \otimes \eta_c \downarrow & & \uparrow e_c \otimes c \\ c \otimes (c^* \otimes c) & \xrightarrow{\cong} & (c \otimes c^*) \otimes c \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_0182 DIA_0182	153	—	—	—	$ \eta_c $
fong-spivak-seven-sketches_0183 DIA_0183	153	—	—	—	$ \epsilon_c $
fong-spivak-seven-sketches_0184 DIA_0184	153	—	—	—	$ c \otimes \eta_c \quad \epsilon_c \otimes c $
fong-spivak-seven-sketches_0185 DIA_0185	153	—	—	—	$ \eta_c \otimes c^* \quad c^* \otimes \epsilon_c $
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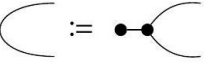
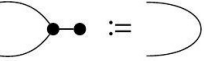
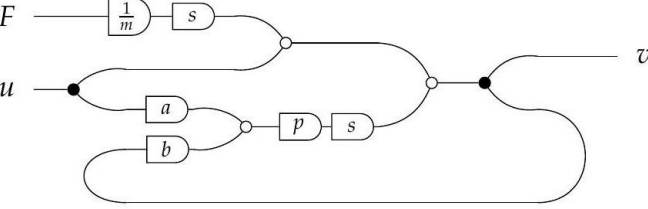
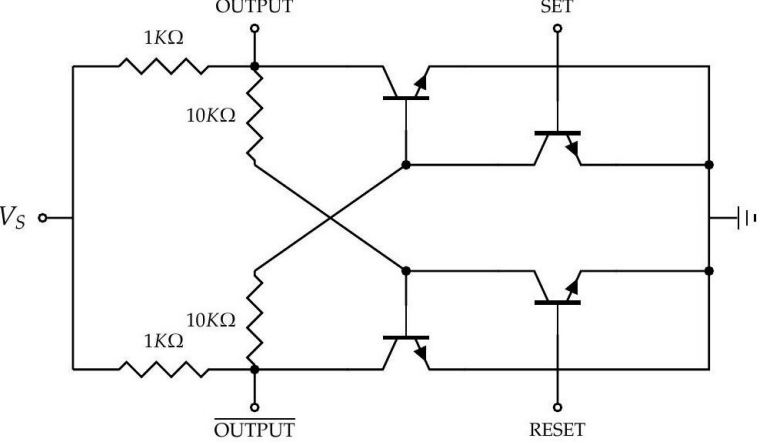
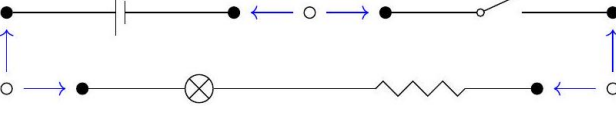
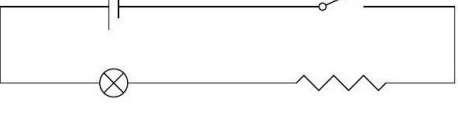
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fong-spivak-seven-sketches_0192	162	—	—	—	
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fong-spivak-seven-sketches_173 DIA_0202	173	—	—	—	
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_177 DIA_0211	177	—	—	—	
fong-spivak-seven-sketches_177 DIA_0212	177	—	—	—	$\begin{array}{c cc} & a & b \\ \hline x & 15 & 3 \\ y & 0 & 21 \end{array} \quad \begin{pmatrix} 15 & 3 \\ 0 & 21 \end{pmatrix}$
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fong-spivak-seven-sketches-DIA_0229	188	—	—	—	
fong-spivak-seven-sketches-DIA_0230	188	—	—	—	
fong-spivak-seven-sketches-DIA_0231	188	—	—	—	
fong-spivak-seven-sketches-DIA_0232	188	—	—	—	
fong-spivak-seven-sketches-DIA_0233	189	—	—	—	
fong-spivak-seven-sketches-DIA_0234	189	—	—	—	
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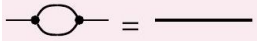
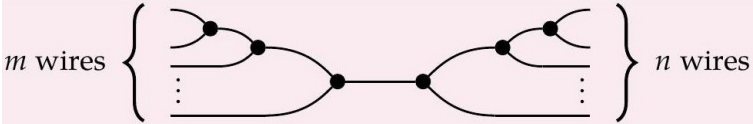
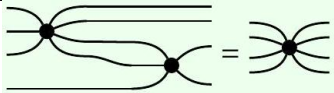
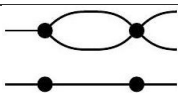
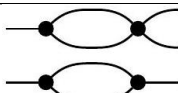
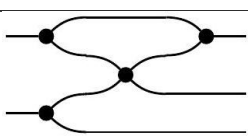
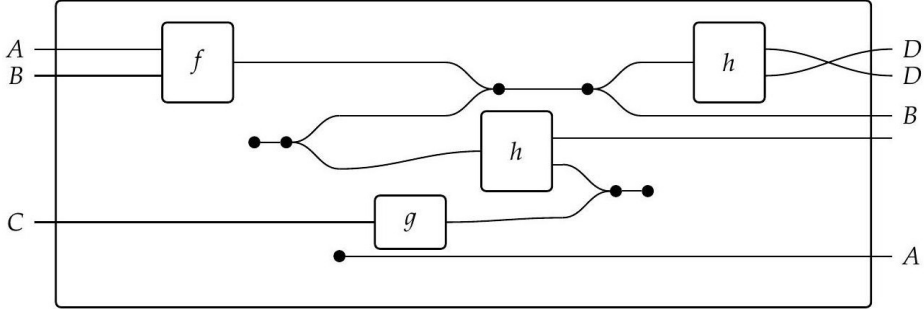
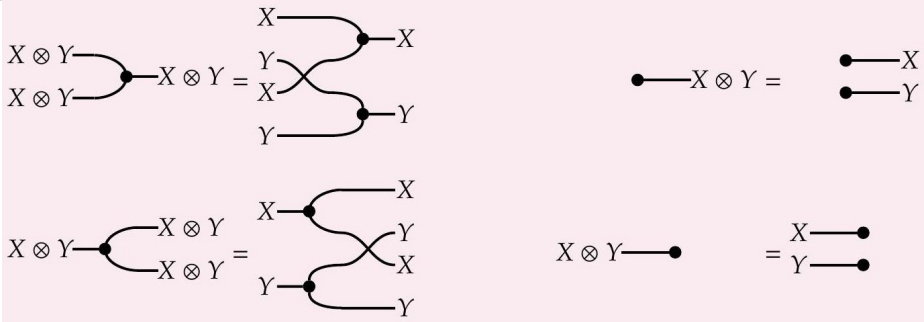
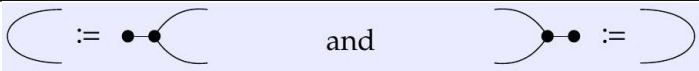
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fong-spivak-seven-sketches_0243	194	—	—	—	
fong-spivak-seven-sketches_0244	194	—	—	—	
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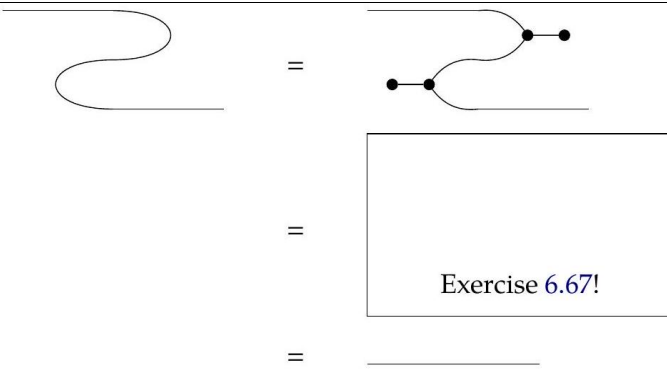
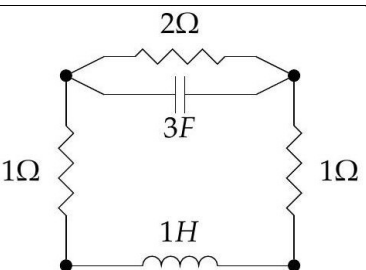
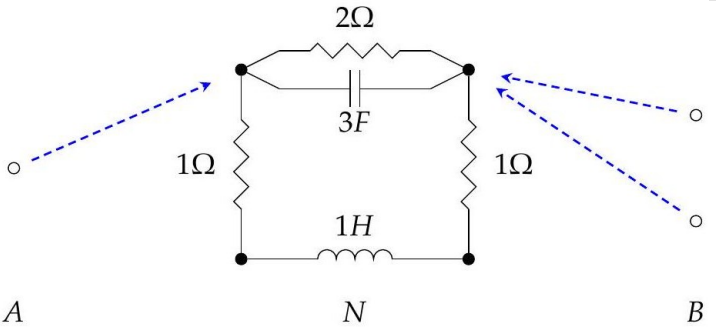
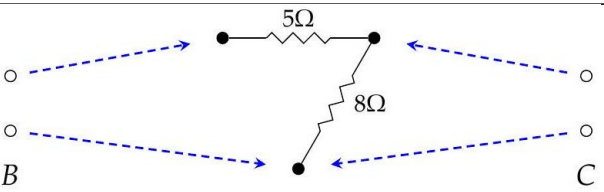
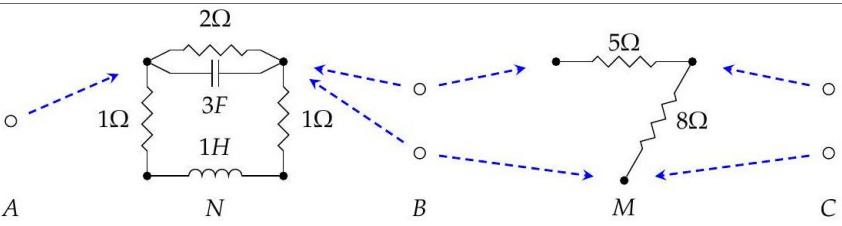
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fong-spivak-seven-sketches_ DIA_0252	200	—	—	—	
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_201 DIA_0256	201	—	—	—	$\begin{array}{ccccc} A & \xrightarrow{f} & X & & \\ i \downarrow & & \downarrow j & \searrow x & \\ A' & \xrightarrow{f'} & X' & & \\ & \searrow a & \swarrow x' & \searrow & \\ & & & & T \end{array}$
fong-spivak-seven-sketches_202 DIA_0257	202	—	—	—	$f: \underline{4} \rightarrow \underline{5}$
fong-spivak-seven-sketches_202 DIA_0258	202	—	—	—	$g: \underline{4} \rightarrow \underline{3}$
fong-spivak-seven-sketches_202 DIA_0259	202	—	—	—	$\begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow \iota_X \\ Y & \xrightarrow{\iota_Y} & X + Y \end{array}$
fong-spivak-seven-sketches_202 DIA_0260	202	—	—	—	$\begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array}$
fong-spivak-seven-sketches_203 DIA_0261	203	—	—	—	$\begin{array}{ccccccc} X & 0 & 1 & 2 & 3 & 4 & 5 & \cdots \\ f \uparrow & \vdots & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\ A & 0 & 1 & 2 & 3 & 4 & 5 & \cdots \\ g \downarrow & \vdots & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \\ Y & 0 & 1 & 2 & 3 & 4 & 5 & \cdots \end{array}$
fong-spivak-seven-sketches_203 DIA_0262	203	—	—	—	$D_1 \rightarrow D_2 \leftarrow D_3 \rightarrow D_4 \rightarrow D_5$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_1 DIA_0297	216	—	—	—	$ \begin{array}{ccc} P(S) \times P(T) & \xrightarrow{\varphi_{S,T}} & P(S \times T) \\ \text{im}_f \times \text{im}_g \downarrow & & \downarrow \text{im}_f \times g \\ P(S') \times P(T') & \xrightarrow{\varphi_{S',T'}} & P(S' \times T') \end{array} $
fong-spivak-seven-sketches_1 DIA_0298	217	—	—	—	
fong-spivak-seven-sketches_1 DIA_0299	217	—	—	—	
fong-spivak-seven-sketches_1 DIA_0300	217	—	—	—	
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fong-spivak-seven-sketches_1 DIA_0305	220	—	—	—	
fong-spivak-seven-sketches_1 DIA_0306	220	—	—	—	
fong-spivak-seven-sketches_1 DIA_0307	220	—	—	—	
fong-spivak-seven-sketches_1 DIA_0308	220	—	—	—	
fong-spivak-seven-sketches_1 DIA_0309	220	—	—	—	$b :=$ and $s :=$
fong-spivak-seven-sketches_1 DIA_0310	221	—	—	—	
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fong-spivak-seven-sketches_1 DIA_0312	221	—	—	—	
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fong-spivak-seven-sketches_0319	222	—	—	—	
fong-spivak-seven-sketches_0320	222	—	—	—	
fong-spivak-seven-sketches_0321	223	—	—	—	
fong-spivak-seven-sketches_0322	223	—	—	—	
fong-spivak-seven-sketches_0323	223	—	—	—	$\eta := \emptyset$
fong-spivak-seven-sketches_0324	223	—	—	—	$\emptyset =: \epsilon$
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_228 DIA_0329	228	—	—	—	
fong-spivak-seven-sketches_228 DIA_0330	228	—	—	—	
fong-spivak-seven-sketches_229 DIA_0331	229	—	—	—	$\varphi :=$
fong-spivak-seven-sketches_229 DIA_0332	229	—	—	—	
fong-spivak-seven-sketches_230 DIA_0333	230	—	—	—	

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fong-spivak-seven-sketches_234 DIA_0334	234	—	—	—	
fong-spivak-seven-sketches_234 DIA_0335	234	—	—	—	
fong-spivak-seven-sketches_236 DIA_0336	236	—	—	—	
fong-spivak-seven-sketches_237 DIA_0337	237	—	—	—	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & & \downarrow & \lrcorner & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$
fong-spivak-seven-sketches_237 DIA_0338	237	—	—	—	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$
fong-spivak-seven-sketches_237 DIA_0339	237	—	—	—	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C' \end{array}$
fong-spivak-seven-sketches_238 DIA_0340	238	—	—	—	$\begin{array}{ccc} A & \xrightarrow{\text{id}_A} & A \\ \text{id}_A \downarrow & \lrcorner & \downarrow f \\ A & \xrightarrow{f} & B \end{array} \quad \begin{array}{ccc} A & \xrightarrow{f} & B \\ f \downarrow & \lrcorner & \downarrow \text{id}_B \\ B & \xrightarrow{\text{id}_B} & B \end{array}$
fong-spivak-seven-sketches_238 DIA_0341	238	—	—	—	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow & \lrcorner & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array}$
fong-spivak-seven-sketches_238 DIA_0342	238	—	—	—	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel & \lrcorner & \parallel \\ A & \xrightarrow{f} & B \end{array}$
fong-spivak-seven-sketches_238 DIA_0343	238	—	—	—	$\begin{array}{ccc} A' & \xrightarrow{g} & A \\ f' \downarrow & \lrcorner & \downarrow f \\ B' & \xrightarrow{h} & B \end{array}$

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fong-spivak-seven-sketches_1 DIA_0344	239	—	—	—	
fong-spivak-seven-sketches_1 DIA_0345	240	—	—	—	
fong-spivak-seven-sketches_1 DIA_0346	240	—	—	—	
fong-spivak-seven-sketches_1 DIA_0347	245	—	—	—	
fong-spivak-seven-sketches_1 DIA_0348	245	—	—	—	
fong-spivak-seven-sketches_1 DIA_0349	245	—	—	—	
fong-spivak-seven-sketches_1 DIA_0350	245	—	—	—	
fong-spivak-seven-sketches_1 DIA_0351	247	—	—	—	

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fong-spivak-seven-sketches_250 DIA_0352	250	—	—	—	
fong-spivak-seven-sketches_251 DIA_0353	251	—	—	—	
fong-spivak-seven-sketches_251 DIA_0354	251	—	—	—	
fong-spivak-seven-sketches_252 DIA_0355	252	—	—	—	
fong-spivak-seven-sketches_252 DIA_0356	252	—	—	—	

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fong-spivak-seven-sketches_0357	253	—	—	—	
fong-spivak-seven-sketches_0358	261	—	—	—	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow & \lrcorner & \downarrow \text{true}^T \\ S & \xrightarrow{p'} & \Omega^T \end{array}$
fong-spivak-seven-sketches_0359	262	—	—	—	$\begin{array}{ccc} \{S \times T \mid p\} & \xrightarrow{\quad} & S \times T \\ \downarrow & & \downarrow \pi_S \\ \exists_t p & \xrightarrow{\quad} & S \end{array}$
fong-spivak-seven-sketches_0360	271	—	—	—	
fong-spivak-seven-sketches_0361	271	—	—	—	$(12) = \begin{array}{c} \bullet \quad * \\ \uparrow \\ \begin{array}{cc} \bullet & * \end{array} \end{array}$ $(1)(2) = \begin{array}{cc} \bullet & * \end{array}$
fong-spivak-seven-sketches_0362	272	—	—	—	
fong-spivak-seven-sketches_0363	273	—	—	—	

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fong-spivak-seven-sketches_273 DIA_0364	273	—	—	—	
fong-spivak-seven-sketches_273 DIA_0365	273	—	—	—	
fong-spivak-seven-sketches_273 DIA_0366	273	—	—	—	
fong-spivak-seven-sketches_274 DIA_0367	274	—	—	—	$\bullet \mapsto p_1$ $* \mapsto p_2$ $\circ \mapsto p_3$
fong-spivak-seven-sketches_275 DIA_0368	275	—	—	—	
fong-spivak-seven-sketches_275 DIA_0369	275	—	—	—	
fong-spivak-seven-sketches_275 DIA_0370	275	—	—	—	

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fong-spivak-seven-sketches_276 DIA_0371	276	—	—	—	
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fong-spivak-seven-sketches_276 DIA_0373	276	—	—	—	
fong-spivak-seven-sketches_278 DIA_0374	278	—	—	—	
fong-spivak-seven-sketches_278 DIA_0375	278	—	—	—	
fong-spivak-seven-sketches_278 DIA_0376	278	—	—	—	
fong-spivak-seven-sketches_278 DIA_0377	278	—	—	—	
fong-spivak-seven-sketches_278 DIA_0378	278	—	—	—	
fong-spivak-seven-sketches_278 DIA_0379	278	—	—	—	
fong-spivak-seven-sketches_279 DIA_0380	279	—	—	—	

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fong-spivak-seven-sketches_279 DIA_0382	279	—	—	—	
fong-spivak-seven-sketches_279 DIA_0383	279	—	—	—	
fong-spivak-seven-sketches_279 DIA_0384	279	—	—	—	
fong-spivak-seven-sketches_279 DIA_0385	279	—	—	—	$c =$
fong-spivak-seven-sketches_279 DIA_0386	279	—	—	—	$g_!(c) =$
fong-spivak-seven-sketches_279 DIA_0387	279	—	—	—	$d =$
fong-spivak-seven-sketches_279 DIA_0388	279	—	—	—	$e =$
fong-spivak-seven-sketches_279 DIA_0389	279	—	—	—	$g^*(d) =$
fong-spivak-seven-sketches_279 DIA_0390	279	—	—	—	$g^*(e) =$
fong-spivak-seven-sketches_280 DIA_0391	280	—	—	—	

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fong-spivak-seven-sketches_280 DIA_0392	280	—	—	—	
fong-spivak-seven-sketches_281 DIA_0393	281	—	—	—	
fong-spivak-seven-sketches_285 DIA_0394	285	—	—	—	
fong-spivak-seven-sketches_285 DIA_0395	285	—	—	—	
fong-spivak-seven-sketches_285 DIA_0396	285	—	—	—	
fong-spivak-seven-sketches_290 DIA_0397	290	—	—	—	
fong-spivak-seven-sketches_290 DIA_0398	290	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_0399 DIA_0399	290	—	—	—	$G_1 = \begin{array}{ccc} & f & \\ \bullet & \xrightarrow{\quad} & \bullet \\ & g & \\ f = g & & \end{array}$ $G_2 = \begin{array}{c} f \\ \bullet \\ \bullet \\ a \\ f = a \end{array}$ $G_3 = \begin{array}{ccc} & f & \\ \bullet & \xrightarrow{\quad} & \bullet \\ g \downarrow & & \downarrow h \\ \bullet & \xrightarrow{\quad} & \bullet \\ f \circ h = g \circ i & & \end{array}$ $G_4 = \begin{array}{ccc} & f & \\ \bullet & \xrightarrow{\quad} & \bullet \\ g \downarrow & & \downarrow h \\ \bullet & & \bullet \\ \text{no equations} & & \end{array}$
fong-spivak-seven-sketches_0400 DIA_0400	291	—	—	—	
fong-spivak-seven-sketches_0401 DIA_0401	291	—	—	—	
fong-spivak-seven-sketches_0402 DIA_0402	291	—	—	—	
fong-spivak-seven-sketches_0403 DIA_0403	292	—	—	—	$\begin{array}{c} s \\ \bullet \\ \bullet \\ z \\ s \circ s = z \end{array}$
fong-spivak-seven-sketches_0404 DIA_0404	292	—	—	—	$\begin{array}{ccccc} a & \xrightarrow{f} & b & \xrightarrow[g]{h} & c \\ f \circ g = f \circ h & & & & \end{array}$
fong-spivak-seven-sketches_0405 DIA_0405	292	—	—	—	
fong-spivak-seven-sketches_0406 DIA_0406	293	—	—	—	$\begin{array}{ccccc} F(c) & \xrightarrow{\alpha_c} & G(c) & \xrightarrow{\beta_c} & H(c) \\ F(f) \downarrow & & G(f) \downarrow & & H(f) \downarrow \\ F(c') & \xrightarrow{\alpha_c} & G(c') & \xrightarrow{\beta_c} & H(c') \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image																
fong-spivak-seven-sketches_293 DIA_0407	293	—	—	—	$\begin{array}{ccc} F(c) & \xrightarrow{\text{id}_{F(c)}} & F(c) \\ F(f) \downarrow & & \downarrow F(f) \\ F(c') & \xrightarrow{\text{id}_{F(c')}} & F(c') \end{array}$																
fong-spivak-seven-sketches_294 DIA_0408	294	—	—	—																	
fong-spivak-seven-sketches_294 DIA_0409	294	—	—	—																	
fong-spivak-seven-sketches_294 DIA_0410	294	—	—	—	Arrow source target Vertex Gr State next DDS <table><tr><th>State</th><th>next</th></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>5</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>5</td></tr><tr><td>6</td><td>7</td></tr><tr><td>7</td><td>6</td></tr></table> Set	State	next	1	4	2	4	3	5	4	5	5	5	6	7	7	6
State	next																				
1	4																				
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6	7																				
7	6																				
fong-spivak-seven-sketches_295 DIA_0411	295	—	—	—																	
fong-spivak-seven-sketches_295 DIA_0412	295	—	—	—	$E \xleftarrow{3} B \xrightarrow{1} G \xrightarrow{2} P \qquad S \begin{matrix} \xrightarrow{4} D \\ \xleftarrow{5} \end{matrix}$ 6																
fong-spivak-seven-sketches_298 DIA_0413	298	—	—	—	(monoid, this book) (preorder, nothing) (category, this book) (monoid, nothing) (preorder, nothing) (category, nothing)																

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fong-spivak-seven-sketches_DIA_0414	300	—	—	—	
fong-spivak-seven-sketches_DIA_0415	301	—	—	—	
fong-spivak-seven-sketches_DIA_0416	302	—	—	—	
fong-spivak-seven-sketches_DIA_0417	302	—	—	—	
fong-spivak-seven-sketches_DIA_0418	303	—	—	—	
fong-spivak-seven-sketches_DIA_0419	303	—	—	—	
fong-spivak-seven-sketches_DIA_0420	305	—	—	—	

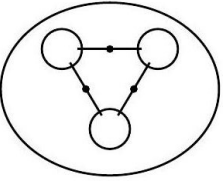
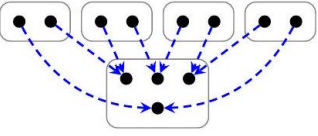
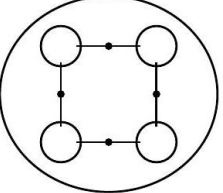
Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_305 DIA_0421	305	—	—	—	
fong-spivak-seven-sketches_305 DIA_0422	305	—	—	—	
fong-spivak-seven-sketches_305 DIA_0423	305	—	—	—	
fong-spivak-seven-sketches_306 DIA_0424	306	—	—	—	
fong-spivak-seven-sketches_306 DIA_0425	306	—	—	—	
fong-spivak-seven-sketches_307 DIA_0426	307	—	—	—	
fong-spivak-seven-sketches_307 DIA_0427	307	—	—	—	
fong-spivak-seven-sketches_307 DIA_0428	307	—	—	—	
fong-spivak-seven-sketches_309 DIA_0429	309	—	—	—	
fong-spivak-seven-sketches_309 DIA_0430	309	—	—	—	
fong-spivak-seven-sketches_310 DIA_0431	310	—	—	—	

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fong-spivak-seven-sketches DIA_0432	310	—	—	—	
fong-spivak-seven-sketches DIA_0433	310	—	—	—	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$
fong-spivak-seven-sketches DIA_0434	310	—	—	—	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$
fong-spivak-seven-sketches DIA_0435	310	—	—	—	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$
fong-spivak-seven-sketches DIA_0436	310	—	—	—	
fong-spivak-seven-sketches DIA_0437	311	—	—	—	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$
fong-spivak-seven-sketches DIA_0438	311	—	—	—	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$
fong-spivak-seven-sketches DIA_0439	311	—	—	—	
fong-spivak-seven-sketches DIA_0440	311	—	—	—	
fong-spivak-seven-sketches DIA_0441	311	—	—	—	
fong-spivak-seven-sketches DIA_0442	311	—	—	—	

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fong-spivak-seven-sketches_311 DIA_0443	311	—	—	—	
fong-spivak-seven-sketches_311 DIA_0444	311	—	—	—	
fong-spivak-seven-sketches_312 DIA_0445	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0446	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0447	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0448	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0449	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0450	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0451	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0452	312	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches DIA_0453	316	—	<pre> \begin{lstlisting}[mathescape=true] \([f, g]: A \sqcup B \longrightarrow T\) apple1 \mapsto \langle a \rangle banana1 \mapsto \langle b \rangle pear1 \mapsto \langle p \rangle cherry1 \mapsto \langle c \rangle orange1 \mapsto \langle o \rangle apple2 \mapsto \langle e \rangle tomato2 \mapsto \langle o \rangle mango2 \mapsto \langle o \rangle. \end{lstlisting} </pre>	(not rendered)	$[f, g]: A \sqcup B \longrightarrow T$ $\text{apple1} \mapsto a$ $\text{banana1} \mapsto b$ $\text{pear1} \mapsto p$ $\text{cherry1} \mapsto c$ $\text{orange1} \mapsto o$ $\text{apple2} \mapsto e$ $\text{tomato2} \mapsto o$ $\text{mango2} \mapsto o.$
fong-spivak-seven-sketches DIA_0454	316	—	—	—	A commutative diagram with nodes A, B, C, and D. Arrows: $A \xrightarrow{\iota_A} A+B \xleftarrow{\iota_B} B$, $A \xrightarrow{f} C$, $B \xrightarrow{g} C$, $C \xrightarrow{h} D$. A central arrow $[f, g]: A+B \rightarrow C$ connects the top node to the middle node.
fong-spivak-seven-sketches DIA_0455	316	—	—	—	A commutative diagram with nodes A, A+B, and B. Arrows: $A \xrightarrow{\iota_A} A+B \xleftarrow{\iota_B} B$. A central arrow $\text{id}_{A+B}: A+B \rightarrow A+B$ connects the top node to the bottom node.
fong-spivak-seven-sketches DIA_0456	317	—	—	—	A complex commutative diagram with nodes A, B, C, D, E, and F. Arrows: $A \xrightarrow{\iota_A} A+B \xleftarrow{\iota_B} B$, $A \xrightarrow{f} C$, $B \xrightarrow{g} D$, $C \xrightarrow{\iota_C} C+D \xleftarrow{\iota_D} D$, $C \xrightarrow{h} E$, $D \xrightarrow{k} F$, $E \xrightarrow{h+k} E+F$. A central arrow $f+g: A+B \rightarrow C+D$ connects the top node to the middle node.
fong-spivak-seven-sketches DIA_0457	318	—	—	—	A diagram illustrating a pushout. It shows two sets of points (represented by dots) at the top, connected by arrows to a single set of points at the bottom. The arrows are labeled f and g. The word "pushout" is written next to the diagram.

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fong-spivak-seven-sketches_318 DIA_0458	318	—	—	—	
fong-spivak-seven-sketches_318 DIA_0459	318	—	—	—	
fong-spivak-seven-sketches_318 DIA_0460	318	—	—	—	
fong-spivak-seven-sketches_318 DIA_0461	318	—	—	—	
fong-spivak-seven-sketches_319 DIA_0462	319	—	—	—	
fong-spivak-seven-sketches_319 DIA_0463	319	—	—	—	
fong-spivak-seven-sketches_320 DIA_0464	320	—	—	—	
fong-spivak-seven-sketches_320 DIA_0465	320	—	—	—	

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fong-spivak-seven-sketches DIA_0476	323	—	—	—	
fong-spivak-seven-sketches DIA_0477	323	—	—	—	
fong-spivak-seven-sketches DIA_0478	323	—	—	—	
fong-spivak-seven-sketches DIA_0479	324	—	—	—	$ \begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ \downarrow h_1 & & \downarrow h_2 & \lrcorner & \downarrow h_3 \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array} $
fong-spivak-seven-sketches DIA_0480	324	—	—	—	$ \begin{array}{c} X \xrightarrow{\quad p \quad} C \\ \downarrow q \quad \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A \xrightarrow{f} B \xrightarrow{g} C \\ \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A' \xrightarrow{f'} B' \xrightarrow{g'} C' \end{array} $
fong-spivak-seven-sketches DIA_0481	324	—	—	—	$ \begin{array}{c} X \xrightarrow{\quad p \quad} C \\ \downarrow q \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A \xrightarrow{f} B \xrightarrow{g} C \\ \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ B' \xrightarrow{g'} C' \end{array} $
fong-spivak-seven-sketches DIA_0482	324	—	—	—	$ \begin{array}{c} X \xrightarrow{\quad p \quad} C \\ \downarrow q \quad \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A \xrightarrow{f} B \xrightarrow{g} C \\ \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A' \xrightarrow{f'} B' \xrightarrow{g'} C' \end{array} $
fong-spivak-seven-sketches DIA_0483	324	—	—	—	$ \begin{array}{c} X \xrightarrow{\quad p \quad} C \\ \downarrow q \quad \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A \xrightarrow{f} B \xrightarrow{g} C \\ \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A' \xrightarrow{f'} B' \xrightarrow{g'} C' \end{array} $

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fong-spivak-seven-sketches DIA_0484	324	—	—	—	
fong-spivak-seven-sketches DIA_0485	325	—	—	—	
fong-spivak-seven-sketches DIA_0486	325	—	—	—	
fong-spivak-seven-sketches DIA_0487	325	—	—	—	
fong-spivak-seven-sketches DIA_0488	325	—	—	—	
fong-spivak-seven-sketches DIA_0489	326	—	—	—	
fong-spivak-seven-sketches DIA_0490	326	—	—	—	
fong-spivak-seven-sketches DIA_0491	326	—	—	—	

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fong-spivak-seven-sketches_3 DIA_0499	331	—	—	—	
fong-spivak-seven-sketches_5 PIC_0001	5	—	—	—	
fong-spivak-seven-sketches_5 PIC_0002	5	—	—	—	
fong-spivak-seven-sketches_2 PIC_0003	29	—	—	—	$\emptyset$
fong-spivak-seven-sketches_3 PIC_0004	317	—	—	—	a) The map $[\text{id}_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \circ \iota_{B+C}], \iota_C \circ \iota_{B+C}]: (A + B) + C \rightarrow A + (B + C)$ is an